QUICK-COMMERCE DELIVERY INTELLIGENCE

An evidence-led operating strategy for delivery-time prediction, late-order risk, demand zones and rider performance

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Project objective
Turn delivery data into operational decisions: estimate delivery time, identify late-order risk, locate high-demand bottleneck zones, and improve rider allocation.

45,593 Approx. source records	**26.29 min** Average delivery time
~14K Late orders shown in report	**30%** Late-order rate shown
4.64 / 5 Average rider rating	**~1K** Riders represented

Important interpretation note: KPI values above are transcribed from the Power BI dashboard screenshots shared in the project conversation and rounded display values. The underlying CSV exports were not available for independent recalculation at report-generation time.

Executive summary

The project develops a delivery-intelligence workflow around four connected business questions: how long an order may take, whether it is likely to miss the assumed service threshold, where demand and delivery delays cluster geographically, and which rider groups may need different operational support.

The Power BI overview shared for this project displays approximately 46K deliveries, an average delivery time of 26.29 minutes, approximately 14K late orders, a late-order rate of 0.30, and an average rider rating of 4.64. These figures indicate that late delivery is a material operational issue in the displayed dataset, while rider rating alone should not be treated as a complete measure of performance.

The analysis plan combines exploratory data analysis, regression, classification, K-Means zone clustering, and rider segmentation. The intended outputs include an interactive prediction application and Power BI pages for executive monitoring, delivery delay analysis, dark-store intelligence, and rider performance.

Management priorities

Priority	Decision	Why it matters
1 · Delivery operations	Use traffic- and weather-aware ETA buffers and allocate rider capacity before peak periods.	The dashboard shows a 30% late-order rate; reducing avoidable delay should be the first operational focus.
2 · Zone planning	Validate high-demand, high-lateness clusters as candidate micro-hub locations.	Geographic concentration can reveal where the current fulfilment footprint may be inefficient.
3 · Rider management	Use rider segments to target coaching, maintenance support and suitable dispatch allocation.	Segment-level performance is more actionable than treating the rider fleet as one group.

How to read the evidence

Observed dashboard values, values embedded in the recommendation-generation code, and proposed actions are kept distinct. The recommendation script contains detailed numeric examples, but the notebook's own final checklist says these fixed findings must be validated against the latest generated CSV exports before being presented as final factual claims. Therefore, zone and rider figures below are labelled as provisional code-reported results, not independently re-audited measurements.

1. Business problem and project scope

Delivery platforms face a trade-off between speed, reliability and rider capacity. Long delivery times can reduce customer satisfaction, while inefficient dispatching can increase workload and operational cost. The project aims to make this trade-off measurable using order, location, traffic, weather and rider-related fields.

The workflow is designed to deliver:

- Delivery-time estimation using regression.
- Late-order risk scoring using classification.
- Geographic demand clustering to identify candidate service zones.
- Rider performance segmentation for targeted operational interventions.
- Exactly three prioritized business recommendations.

2. Data and preparation

The project brief describes a source dataset of approximately 45,593 delivery records and 20 columns. The data-cleaning workflow is intended to standardize delivery-time values, convert age/rating/delivery-count fields to numeric types, clean category labels, inspect missing values and duplicates, validate GPS coordinates, and document imputation decisions.

Area	Treatment / control	Business reason
Delivery time target	Parse values such as “(min) 24” into numeric minutes.	Ensures regression and SLA calculations use a numeric target.
GPS coordinates	Check valid latitude/longitude ranges and calculate Haversine distance.	Supports distance analysis and geographic clustering.

Missing values	Apply documented, field-appropriate imputation rather than deleting records indiscriminately.	Retains useful observations while making assumptions visible.
Categorical fields	Normalize whitespace and inconsistent category labels.	Prevents duplicate categories and inconsistent model inputs.
Leakage prevention	Exclude IDs, the target, rider ID as a direct feature, and information unavailable at order placement; do not use pickup time for order-placement predictions.	Keeps evaluation closer to real deployment conditions.

The modelling design uses a held-out test split and fits preprocessing inside pipelines on training data. This is important because fitting imputers, scalers or encoders on the full dataset can leak information from the test data into model development.

3. Exploratory analysis and operational signals

The planned EDA covers delivery-time distribution; delivery time by weather, traffic, hour, distance, vehicle and city type; late-order rate by weather and traffic; and rider rating versus delivery time. These views are intended to identify operational associations, not prove causation.

The Power BI pages already created provide an executive overview, delivery delay analysis, dark-store intelligence and rider performance. The executive page's visible KPIs are summarized below.

Dashboard indicator	Displayed value	Interpretation
Total deliveries	**~46K**	Rounded display; aligns with the project brief's approximate dataset size.
Average delivery time	**26.29 min**	Baseline operational delivery duration in the dashboard.
Late orders	**~14K**	A sizeable number of deliveries exceed the configured late threshold.
Late-order rate	**0.30 (30%)**	The displayed decimal corresponds to approximately 30%.
Average rider rating	**4.64**	High average rating does not rule out delay issues at zone or traffic level.

4. Machine-learning approach

Delivery-time regression

The model pipeline compares a mean baseline, Ridge/Linear Regression, Random Forest, Gradient Boosting and XGBoost when available. Evaluation is defined using test MAE, RMSE and R^2. The project code selects the model with the lowest test MAE in its comparison workflow; the final report should include the generated comparison CSV values before quoting a numerical model score.

Late-order classification

The classifier predicts whether delivery time exceeds a configurable SLA threshold. Candidate models include Logistic Regression, Random Forest and Gradient Boosting, with XGBoost optional. Evaluation includes precision, recall, F1-score, ROC-AUC, confusion matrix and ROC curve. The challenge brief does not supply a fixed SLA number, so the chosen threshold must be stated as a project assumption rather than an official benchmark.

Model results status

The available notebook and export script describe the evaluation process and expected CSV outputs, but the actual generated model-comparison tables were not present in the files retrieved for this report. No MAE, RMSE, R^2, ROC-AUC, F1 or recall values have been fabricated. Insert the final values from reports/model_comparison_regression.csv and reports/model_comparison_classification.csv after confirming the notebook ran successfully.

5. Geographic demand and dark-store intelligence

The project uses K-Means clustering over valid geographic coordinates, with scaled coordinates, an elbow-method check and silhouette evaluation. For each cluster, the intended summary includes order demand, demand share, average delivery time, late-order rate, average distance and dominant city.

The recommendation-generation code identifies Zone 1 and Zone 6 as candidate bottleneck zones. Its embedded figures are:

Candidate zone	Orders	Late rate	Avg. delivery time
Zone 1	5,848	31.65%	26.57 min
Zone 6	6,326	29.97%	Not stated in the retrieved recommendation text

The same recommendation code states that the two zones together represent 29.0% of valid deliveries and 34.2% of network SLA delays, with an average order distance of 10.4 km. These are provisional code-reported figures and must be checked against the current zone_clusters.csv export before external presentation.

Business interpretation

High order concentration combined with above-network late rates can indicate a candidate location for a micro-fulfilment hub or a route-capacity intervention. However, this dataset is a food-delivery dataset: restaurant coordinates are proxies, not confirmed quick-commerce dark-store locations. The coordinates should therefore guide a feasibility study, not be treated as a final real-estate decision.

Recommended next validation

Compare each cluster with the network-wide average; verify coordinate quality and city context; then validate shortlisted locations using actual customer demand, property availability, rent, inventory replenishment, road access and service-radius analysis.

6. Rider performance segmentation

Riders are aggregated using delivery volume, average delivery time, average rating, late-order rate and vehicle condition where available. K-Means groups are interpreted as performance tiers rather than permanent labels for individual riders.

The recommendation script reports the following provisional segment comparison:

Segment	Riders	Late rate	Avg. time	Rating	Vehicle condition
Tier 1 · Elite	349	20.65%	24.61 min	4.69	1.8
Tier 3 · At-risk	335	39.68%	28.19 min	4.60	0.8

The code reports Tier 3 as 25.38% of the rider fleet. The comparison suggests a gap between the two groups in late-order rate, delivery time, rating and vehicle condition. These figures must be reconciled with rider_segments.csv before being treated as final measured results.

7. Three prioritized business recommendations

Recommendation 1 — Dynamic traffic-aware ETA and peak rider pre-allocation

Priority: P1 · Critical **Domain:** Delivery operations **Timeline:** Weeks 1–4

Finding and evidence: The recommendation code reports average delivery time of 31.14 minutes and a 51.4% late rate under Jam traffic, compared with 21.41 minutes and an 11.2% late rate under Low traffic. It also reports that traffic density and weather contribute over 25% of tree-model predictive importance.

Action: Use the trained ETA model to adjust customer-facing delivery estimates during severe traffic and weather. Pilot pre-allocation of additional floating riders before evening peaks, then measure late-rate, ETA error and rider utilization against a comparable control period. The proposed 25% additional rider allocation is a pilot hypothesis, not a proven uplift.

Recommendation 2 — Validate Zone 1 and Zone 6 as micro-fulfilment candidates

Priority: P2 · High **Domain:** Zone planning **Timeline:** Months 2–4

Finding and evidence: The recommendation code reports 5,848 orders and a 31.65% late rate for Zone 1, and 6,326 orders with a 29.97% late rate for Zone 6. It further reports a combined 29.0% share of valid deliveries and 34.2% of network SLA delays.

Action: Conduct a feasibility assessment for micro-hubs or smaller fulfilment points in these zones. Validate actual demand, road access, property cost and replenishment requirements before selecting a site. Do not interpret the model centroids as verified property locations.

Recommendation 3 — Tiered rider coaching, equipment checks and dispatch support

Priority: P3 · Medium **Domain:** Rider management **Timeline:** Weeks 2–8

Finding and evidence: The recommendation code reports a 39.68% late rate and 28.19-minute average delivery time for Tier 3, compared with 20.65% and 24.61 minutes for Tier 1. The reported average vehicle-condition scores are 0.8 and 1.8 respectively.

Action: Offer navigation coaching, vehicle-condition checks and maintenance support to at-risk riders. Test dispatch rules that reserve complex or high-priority orders for riders with demonstrated reliability, while tracking fairness and rider workload.

8. Limitations and assumptions

- **SLA threshold:** The challenge brief does not prescribe a fixed numerical SLA. The late-order definition depends on the configurable project threshold and must be stated in the final submission.
- **Proxy geography:** The source is a food-delivery dataset. Restaurant/customer coordinates are not direct evidence of ideal quick-commerce dark-store sites.
- **Recommendation figures:** The recommendation script embeds fixed numeric findings. The notebook checklist explicitly instructs the team to validate them against the latest generated CSVs before use.
- **Model performance:** Actual model-comparison CSVs were not available among the retrieved files for this report. Numerical model scores are intentionally not included.
- **Observational data:** Associations between traffic, weather, distance, rider attributes and delivery time do not establish causation.
- **Operational estimates:** Any proposed rider uplift, ETA buffer or expected time saving is a pilot target until measured in a controlled operational trial.

9. Conclusion

The Quick-Commerce Delivery Intelligence project establishes a practical analytics framework for monitoring delivery performance, estimating ETA, identifying late-order risk, prioritizing geographic bottlenecks and supporting rider development. The dashboard indicates a material late-delivery challenge, while the zone and rider analyses provide targeted hypotheses for operational improvement. The next step is to validate every provisional finding against the latest exported datasets and complete the report with held-out model metrics before final judging or business use.

Appendix — Evidence and files to verify before submission

Artifact	What to confirm
powerbi/cleaned_data.csv	Final row count, late-order definition, valid coordinates and KPI totals.
powerbi/model_metrics.csv	Regression and classification test metrics; confirm selected champions.
powerbi/zone_clusters.csv	Cluster demand, late rate, average delivery time, distance and centroids.
powerbi/rider_segments.csv	Rider counts and segment-level delivery, rating, late-rate and vehicle metrics.
powerbi/business_findings.csv	Exactly three recommendations and all numerical evidence matched to current outputs.
reports/model_comparison_*.csv	Complete held-out test comparison tables for regression and classification.

Reference basis: the uploaded sample business report “Catch Fraud, Not Customers” was used as the editorial/layout reference. Project scope and methodology were taken from the Quick-Commerce project brief and notebook; KPI values were transcribed from the user's Power BI screenshots; provisional zone and rider findings were taken from the recommendation-generation code.

Before submission: rerun the notebook, compare all figures with the exported CSVs, and replace any provisional values that differ.